HARISHALINEE ELANGOVAN

Software, Data & AI Professional | .NET | Python | SQL | Automation Testing
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PROFESSIONAL SUMMARY

Associate Data Analyst & AI Engineer and BSc (Hons) Information Technology (Data Science) undergraduate at SLIIT, with hands-on experience in full-stack .NET Framework development, REST APIs, IIS deployment, Docker, OCR and document processing, automation testing, data pipelines, interactive dashboards and machine learning. Built an edge-based fall-risk detection model with 97.12% accuracy on unseen subjects and co-authored a paper accepted for ICSCDS-2026 (IEEE proceedings). Proficient in Python, SQL, Power BI, TensorFlow, PyTorch, AWS and Azure. Open to Software Engineer, Data Analyst, AI/ML Engineer and QA Automation roles.

TECHNICAL SKILLS

- **Core Competencies:** Software development, data analytics, business intelligence, data visualization, data pipelines, ETL, machine learning, deep learning, NLP, computer vision, test automation, application deployment, containerization, OCR, document processing, cloud computing
- **Programming & Frameworks:** Python, SQL, .NET Framework (back-end and front-end), REST APIs, Java, C/C++, R, FastAPI, Flask, React
- **Data Analysis & BI:** Power BI, SSRS, Advanced Excel, Matplotlib, Seaborn, Streamlit, Monday.com, ETL, SSIS, data warehousing, OLAP (SSAS), data modeling, Oracle, SQL Server
- **AI & Machine Learning:** Machine learning, deep learning, Scikit-Learn, XGBoost, TensorFlow, PyTorch, NLP, computer vision (MediaPipe), LangChain, LangGraph, Gemini API, generative AI (StyleGAN)
- **Cloud, Testing & Deployment:** AWS, Microsoft Azure, IIS, Docker, application deployment, automation testing, Git, GitHub
- **Professional:** Problem-solving, stakeholder management, information systems auditing, controls and risk assessment

WORK EXPERIENCE

Associate Data Analyst & AI Engineer | Agrithmics | Feb 2026 – Present
- Develop back-end and front-end features in .NET Framework and build REST APIs
- Deploy applications on IIS, work with Docker containers, and release data/AI solutions to live environments
- Build OCR-based text extraction and document-handling features
- Carry out automation testing to verify that dashboards, data pipelines and reporting workflows work correctly
- Led development of multiple interactive dashboards for plantation performance tracking across 6–7 projects
- Optimized data pipelines that feed AI model training, improving processing efficiency
- Built automated reporting workflows that speed up decision-making

Data Analytics & AI Development Intern | Agrithmics | Nov 2025 – Jan 2026
- Designed interactive dashboards to visualize plantation performance
- Preprocessed and structured large datasets for AI model training and optimization

Data Analyst Intern | International Youth Alliance for Peace (IYAP) | Mar 2025 – Oct 2025
- Used Monday.com to manage data, build dashboards and track project performance
- Analysed data and produced reports to support strategic planning

IT Operator | Crown Global International Pvt Ltd | Jul 2024 – Feb 2025
- Managed digital record-keeping systems and ensured accuracy of computerized accounts
- Performed system maintenance and supported branding operations through media coordination

Freelance Tutor | Self-employed | Jan 2023 – Present
- Provide personalized tutoring sessions that improve students' understanding and academic performance

PROJECTS

Edge-Enabled Skeletal Movement Analysis System for Real-Time Fall Risk Detection | Final-year research (CareSense group project R26-DS-010) | 2026
- Built a privacy-preserving fall-risk detector that runs on an ordinary RGB webcam and a CPU, analysing skeleton data only
- Fused an ST-GCN (PyTorch, MediaPipe) with 18 biomechanical features in a late-fusion model, trained on NTU RGB+D and UR Fall Detection (578 labelled clips)
- Achieved **97.12% accuracy, 92.31% sensitivity, 1.28% false-alarm rate and ROC-AUC 0.9836** on unseen subjects, with 4.5 ms inference per window on a CPU
- Measured domain shift (cross-dataset ROC-AUC 0.410) and recovered to macro-F1 0.8851 on webcam data using 35 fine-tuning clips

Tourism AI – Multi-Agent Travel Optimization System | Python, Gemini API, LangGraph, LangChain, FastAPI, React

- Engineered a multi-agent system that builds personalized, Sri Lanka-aware travel itineraries
- Designed specialized agents for interest understanding, day planning, weather-aware re-planning, transport cost and duration, train options, food and cultural tips
- Delivered as a scalable FastAPI and React full-stack application

Ride-Hailing AI – Data Mining and Prediction System | Python, Scikit-Learn, Streamlit

- Mined about 148,000 ride bookings using Apriori / FP-Growth route mining and K-Means / DBSCAN rider segmentation
- Built models to predict high-demand windows and forecast monthly spend, and recommended money-saving subscription plans in an interactive Streamlit dashboard

Customer Churn Prediction – Telco Churn Lab | Python, Scikit-Learn, XGBoost, Flask, React, Vite

- Trained and compared Random Forest and XGBoost classifiers on 7,043 customers using accuracy, precision, recall and F1; Random Forest performed best
- Shipped a full-stack app with a Flask prediction API and a React + Vite front end

Hotel Booking Data Warehouse and BI | SQL Server, SSIS, SSAS, Power BI, SSRS

- Designed a star schema with a booking fact table and six dimensions, built SSIS ETL packages and an SSAS OLAP cube
- Reported booking trends, revenue and cancellation rates in Power BI and SSRS dashboards

StyleGAN and StyleGAN2 – Bias and Fairness Analysis | PyTorch, StyleGAN2-ADA, OpenCV

- Analysed how StyleGAN2 models trained on FFHQ favour lighter skin tones and Western faces, and proposed a South Asia-aware fairness framework using ITA skin-tone metrics

RESEARCH PUBLICATION

Multimodal AI Framework for Intelligent Caregiver Support and Patient Safety in Care Facilities
Dayana Priyadharshani Kumar, Harishalinee Elangovan, Vidursha Prabagaran, Kaushalya Nagenthiraraja, Samadhi Rathnayake, Samantha Thelijjagoda. 4th International Conference on Sustainable Computing and Data Communication Systems (ICSCDS-2026), IEEE conference proceedings, 15–17 December 2026, India. Status: accepted for presentation and publication (acceptance with major revision), Paper ID ICSCDS-0291.
CareSense research portfolio (project documents, results and presentations): caresense-portfolio.vercel.app

EDUCATION

BSc (Hons) in Information Technology, specializing in Data Science | SLIIT | Jan 2025 – Mar 2027 (expected) | CGPA 3.25 / 4.00

Higher Diploma in Information Technology | SLIIT | Oct 2022 – Nov 2024

Cambridge Advanced Level | Lyceum International School, Sri Lanka | 2020 – 2022 | 1 A*, 1 A, 1 C

Cambridge Ordinary Level | Lyceum International School, Sri Lanka | 2010 – 2020 | 2 A*, 2 A, 2 B, 1 C

CERTIFICATIONS AND COURSES

- AI/ML Engineer (Stage 1 and 2) – The Centre for Open and Distance Education, Faculty of Computing, SLIIT
- AWS Cloud Computing Certifications – AWS
- Microsoft Azure Cloud Certifications – Microsoft
- Information Systems Auditing, Controls and Assurance – Coursera
- Oracle Database 19c; Introduction to Artificial Intelligence; R for Data Science: Analysis and Visualization; Python Data Analysis; Getting Started with Power BI – LinkedIn Learning

PERSONAL DETAILS

Date of birth: 6 October 2004 | Nationality: Sri Lankan

LANGUAGES

English, Tamil, Sinhala

REFERENCES

Mrs Laurel Mathew, Director, Crown Global International (Pvt) Ltd | 0777788969 | mathewlaurel@gmail.com
Dr. Samantha Rajapaksha, Head of IT Department and Senior Lecturer (HG), SLIIT | 0117544802 | samantha.r@sliit.lk